

Database Systems	
Lab #:	01
Lab Title:	Introduction to database and tool usage.
Submitted by:	
Names	Registration #
Sumaira Safeer	FA22-BCE-019

Rubrics name & number					Marks	
					In-Lab	Post-Lab
Engineering Knowledge	R2: Use of Engineering Knowledge and follow Experiment Procedures: Ability to follow experimental procedures, control variables, and record procedural steps on lab report.					
Problem Analysis	R6: Experimental Data Analysis: Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.					
Design	R8: Best Coding Standards: Ability to follow the coding standards and programming practices.					
Modern Tools Usage	R9: Understand Tools: Ability to describe and explain the principles behind and applicability of engineering tools.					
Individual and Teamwork	R12: Individual Work Contributions: Ability to carry out individual responsibilities.					
	R13: Management of Team Work: Ability to appreciate, understand and work with multidisciplinary team members.					
Rubrics #	R2	R6	R8	R9	R12	R13
In Lab						
Post- Lab						

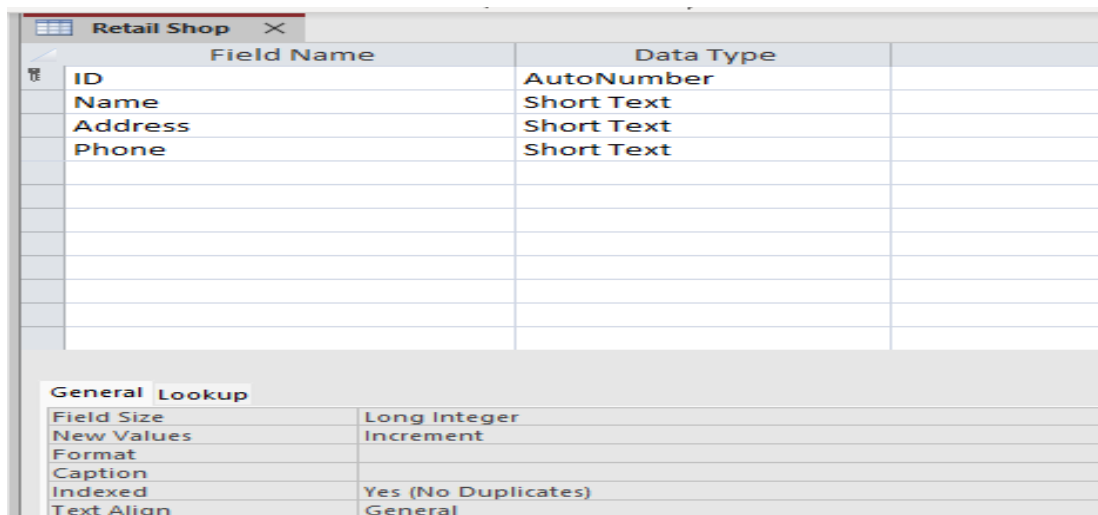
Lab No: 01

Introduction to database and tool usage.

➤ Question No: 01

Develop Retail shop database in MS Access that consist the information of customers (ID, Name, Address, Phone) .

Design View:



The screenshot shows the Design View of a table named 'Retail Shop' in Microsoft Access. The table has four fields: ID (AutoNumber), Name (Short Text), Address (Short Text), and Phone (Short Text). The 'General' tab is selected in the property sheet at the bottom, showing settings for Field Size, New Values, Format, Caption, Indexed (Yes (No Duplicates)), and Text Align (General).

Field Name	Data Type
ID	AutoNumber
Name	Short Text
Address	Short Text
Phone	Short Text

General	
Field Size	Long Integer
New Values	Increment
Format	
Caption	
Indexed	Yes (No Duplicates)
Text Align	General

Datasheet View:

Sort & Filter					Records					Find				
Filter	A↓ Ascending	Selection	Refresh	New	Σ Totals	Find	Replace	Go To	Select					
	Z↓ Descending	Advanced	All	Save	abc Spelling									
	Remove Sort	Toggle Filter	Delete	More										

ID	Name	Address	Phone
1	Rohid	Karachi, Sindh	0308-6795375
2	Aazaan Zarsham	Islamabad, Punjab	0300-4852694
3	Zoha Malik	Faislabad, Punjab	0342-3684253
4	Saqib Hassan	D.I Khan, Khyber Pakhtunkhawa	0333-7654313
5	Rohit Ali	Lahore, Punjab	0302-6348642
6	Aliya Khan	Quetta, Balochistan	0307-2468254
7	Hamza Bilal	Mianwali, Punjab	0344-5732619
8	Hoorain	Rawalpindi, Punjab	0305-8523456
9	Hira Awan	Faislabad, Punjab	
(New)			

- **Searching the information of customer through their address in Retail Shop Query.**

All Access Objects					Retail Shop Query1				
Search...					Retail Shop				
Tables					* ID				
Retail Shop					Name				
Queries					Address				
Retail Shop Query1					Phone				

Field:	ID	Name	Address	Phone
Table:	Retail Shop	Retail Shop	Retail Shop	Retail Shop
Sort:				
Show:	☒	☒	☒	☒
Criteria:			"Faislabad, Punjab"	
or:				

Information as output:

All Access Obj... <

Search...

Tables

Retail Shop

Queries

Retail Shop Query1

ID	Name	Address	Phone
3	Zoha Malik	Faislabad, Punjab	0342-3684253
9	Hira Awan	Faislabad, Punjab	
*	(New)		

- **Searching the information of customer through their Name in Retail Shop Query.**

Retail Shop				
* ID Name Address Phone				
Field:	ID	Name	Address	Phone
Table:	Retail Shop	Retail Shop	Retail Shop	Retail Shop
Sort:				
Show:	☒	☒	☒	☒
Criteria:		Azaan Zarsham		
or:				

Information as output:

All Access Obj... <				
Search...				
Tables				
Retail Shop				
Queries				
Retail Shop Query1				

ID	Name	Address	Phone
2	Aazaan Zarsham	Islamabad, Punjab	0300-4852694
*	(New)		

- Searching the information of customer through their ID in Retail Shop Query then press control+w and then select yes .

All Access Obj... <				
Search...				
Tables				
Retail Shop				
Queries				
Retail Shop Query				


```
SELECT *
FROM Retail AS Shop
WHERE ID=6;
```

Information as output:

All Access Obj... <				
Search...				
Tables				
Retail Shop				
Queries				
Retail Shop Query1				

ID	Name	Address	Phone	Click to Add
1	Rohid	Karachi, Sindh	0308-6795375	
2	Aazaan Zarsham	Islamabad, Punjab	0300-4852694	
3	Zoha Malik	Faislabad, Punjab	0342-3684253	
4	Saqib Hassan	D.I Khan, Khyber Pakhtunkhawa	0333-7654313	
5	Rohit Ali	Lahore, Punjab	0302-6348642	
6	Aliya Khan	Quetta, Balochistan	0307-2468254	
7	Hamza Bilal	Mianwali, Punjab	0344-5732619	
8	Hoorain	Rawalpindi, Punjab	0305-8523456	
9	Hira Awan	Faislabad, Punjab		
*	(New)			

Question No: 02

In-lab Task:

Task No: 01

- Create a database named COMSATS
- Create three tables named Student, Faculty, Course.
- Insert 4 records in each table.

➤ Student:

Design View:

Field Name	Data Type	Description (Optional)
ID	AutoNumber	
Reg No	Short Text	001
First Name	Short Text	Peter
Last Name	Short Text	Lee
Email	Long Text	Plee@university.edu
Date of Birth	Short Text	1992
Department	Short Text	Electrical Engineering
Semester	Short Text	VI

Field Properties	
General	Lookup
Field Size	Long Integer
New Values	Increment
Format	
Caption	
Indexed	Yes (No Duplicates)
Text Align	General

Datasheet View:

student						
Reg No	First Name	Last Name	Email	Date of Birth	Department	Semester
001	Peter	Lee	Plee@university.edu	1992	Electrical Engineering	VI
005	Jonathan	Edwards	Jewards@university.edu	1996	Computer Engineering	V
007	Merilyn	Johnson	Mjohnson@university.edu	2001	Computer Engineering	VII
010	David	Letty	Dletty@university.edu	1886	Electrical Engineering	VIII
*						

➤ Faculty:

Design View:

Field Name	Data Type	Description (Optional)
Name	Short Text	
Department	Short Text	
Email	Short Text	

Datasheet View:

ID	Name	Department	Email
D_123	Haley	Electrical Engineering	Haley@university.edu
D_456	Kim	Computer Engineering	Kim@university.edu
D_932	Martine	Electrical Engineering	Martinez@university.edu
D_987	Mfume	Computer Engineering	Mfume@university.edu
*			

Design View:

[illegible]

Datasheet View:

 Filter
  Ascending
  Descending
  Remove Sort
  Selection
  Advanced
  Toggle Filter
  Refresh All

Sort & Filter

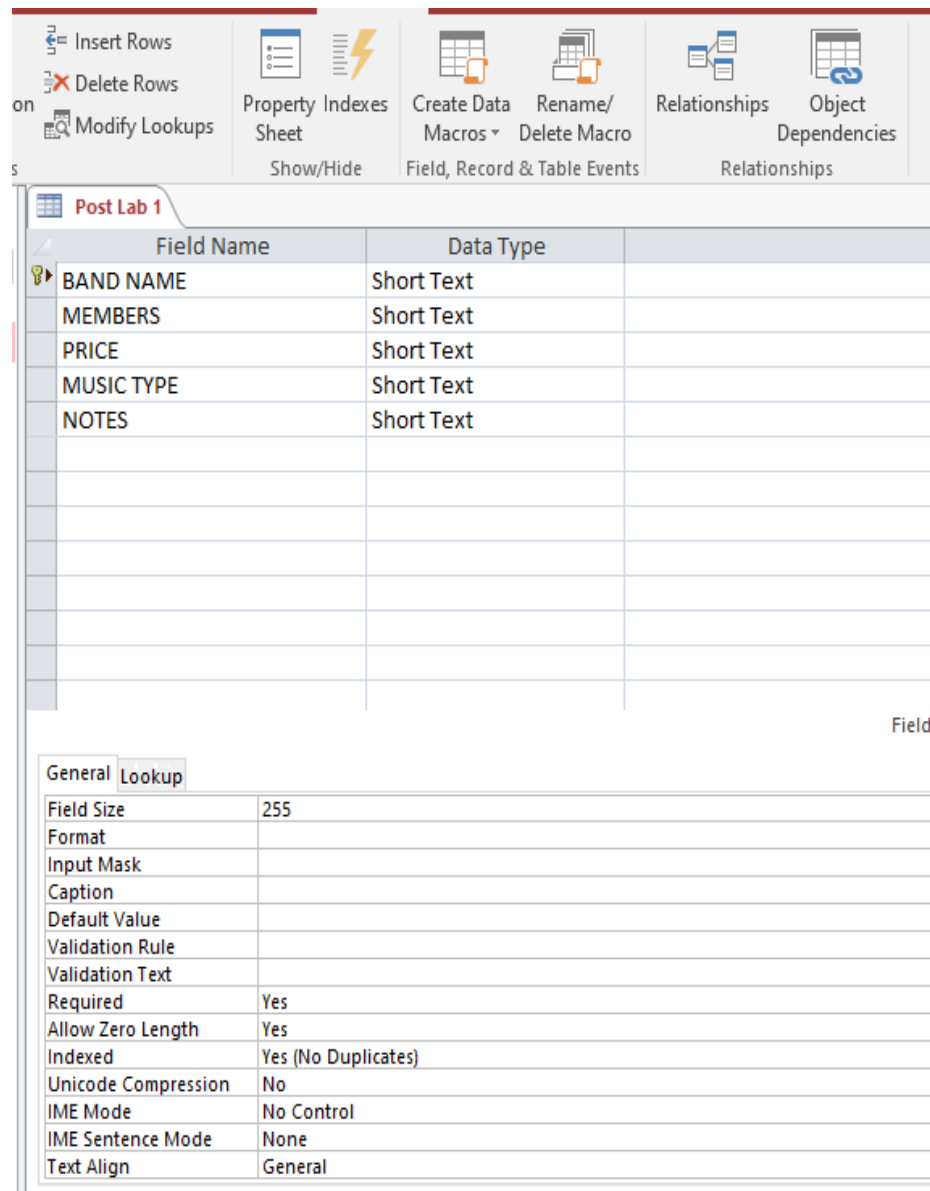
Course	
Course Code	Course Name
CSC112	Data Structure
EEE212	Electronics
EEE414	Data Communication
MTH444	Numerical Computation
*	

Post Lab:

Task No: 01

Create a database file called Bands and table.

Design View:



Field Name	Data Type
BAND NAME	Short Text
MEMBERS	Short Text
PRICE	Short Text
MUSIC TYPE	Short Text
NOTES	Short Text

Field
BAND NAME
MEMBERS
PRICE
MUSIC TYPE
NOTES

General	Lookup
Field Size	255
Format	
Input Mask	
Caption	
Default Value	
Validation Rule	
Validation Text	
Required	Yes
Allow Zero Length	Yes
Indexed	Yes (No Duplicates)
Unicode Compression	No
IME Mode	No Control
IME Sentence Mode	None
Text Align	General

Datasheet View:

Filter Ascending Descending Remove Sort Selection Advanced Toggle Filter Refresh All New Save Delete More Totals Spelling Find Go To Select Calibri (Detail) 11 B I U Sort & Filter Records Find Text Formatting				
Post Lab 1				
BAND NAME	MEMBERS	PRICE	MUSIC TYPE	NOTES
AMTTHEW AND SON	3	750	ROCK & ROLL	a young band, prefers gigs with a younger audience
BLUE SOX QUARTET	4	400	JAZZ	jazz festivals, dances. Prefers smaller gigs.
CRUISING	5	500	JAZZ	suitable for older audiences, has a good repertoire of older dance numbers
HIGH IN DRY	4	500	ROCK & ROLL	weddings, balls, dances preferred
SQUARE DANCE TRIO	3	650	COUNTRY & WE	prefers C&W gigs and concerts suitable for weddings,
YOUNG AND PROUD	5	750	ROCK & ROLL	young band, suits conserts and school functions
*				